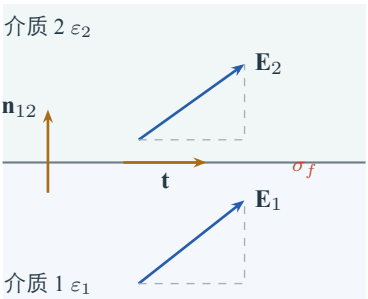
理论图 T31　介质边界条件：统一法向，分别统计电荷

示意取切向、法向分量均正的情形



法向统一由介质 1 指向介质 2；本图为无理想偶极层的普通界面。

$$\mathbf{n}_{12} \times (\mathbf{E}_2 - \mathbf{E}_1) = 0$$

切向电场连续

$$\mathbf{n}_{12} \cdot (\mathbf{D}_2 - \mathbf{D}_1) = \sigma_f$$

法向电位移只由自由面电荷跃变

$$\varphi_2 = \varphi_1$$

普通单层面电荷不使电势跃变

$$\sigma_b = (\mathbf{P}_1 - \mathbf{P}_2) \cdot \mathbf{n}_{12}$$

$$\varepsilon_0(E_{2n} - E_{1n}) = \sigma_f + \sigma_b$$